

CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 08

Name (Print): _____ Section: _____

1. **Strong Dependency**

A **strong dependency** exists when one class directly depends on the _____ of another class.

2. **Strong or Weak dependency?**

_____: When code repeatedly calls many getters on another object.

3. **Association vs. Dependency (UML)**

- **Association** (solid line): an object _____ another object as a field.
- **Dependency** (dashed arrow): one class _____ another temporarily.

4. **Message Chain**

A message chain occurs when a client makes a _____ of method calls to reach and operate on another object.

5. **What is the Message Chain Problem**

One problem caused by message chains is that

_____.

6. **Coupling**

Coupling describes how _____

7. **Cohesion**

Cohesion describes how _____

8. **Cohesion Level**

- **High cohesion** means a class has _____.
- **Low cohesion** means a class has _____.

9. High or Low Coupling

- 1) _____: Each component is as independent as possible
- 2) _____: A change in one class is likely to affect many others

10. Exam question: Which of the following is the best design principle?

- a. Low coupling / High cohesion.
- b. Low coupling / Low cohesion.
- c. High coupling / Low cohesion.
- d. High coupling / High cohesion

11. Exam question: Which of the following are characteristics of encapsulation? (Select all that apply)

- a. Encapsulation hides the internal state of an object.
- b. Encapsulation allows direct access to object fields from outside the class.
- c. Encapsulation is achieved using private fields and public getter/setter methods.
- d. Encapsulation leads to increased coupling

12. Exam question: Which of the following are characteristics of dependency in object-oriented design? (Select all that apply)

- a. A class relying on another class for its functionality.
- b. A change in one class can potentially affect another class.
- c. Dependencies can always be resolved by making all classes public.
- d. Dependencies can be minimized using design patterns

13. Before You Leave

Write one question you still have about today's topic.
